MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
REGULAR EXAMINATION March 2026
Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFSC200 - FORENSICSEROLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks:50**

Part A: Short Answer Questions

Answer any 10. Each carries 2 marks.

1. **Antigen-Antibody Reaction:** A specific chemical interaction between antibodies produced by B cells and antigens during an immune response.
2. **Cross-over Electrophoresis:** Justified as a confirmatory test because it uses an electric field to force antigens and antibodies toward each other, creating a visible precipitin line that confirms species origin with high sensitivity.
3. **Confirmatory Test for Blood: Takayama Test** (Haemochromogen crystal test) or **Teichmann Test** (Haemin crystal test).
4. **Blood Grouping from Dried Stains:** The process of identifying ABO antigens or antibodies in desiccated blood using techniques like **Absorption-Elution** or **Absorption-Inhibition**.
5. **Significance of Blood Grouping:** Used for the exclusion of suspects, linking a victim/suspect to a scene, and resolving disputed paternity.
6. **Rouleaux Formation:** The stacking of red blood cells like a "pile of coins," often occurring in thick blood smears or due to certain plasma protein imbalances, which can interfere with grouping.
7. **Kastle-Mayer Procedure:** 1. Rub a moist swab on the stain. 2. Add a drop of KM reagent (phenolphthalein). 3. Add a drop of Hydrogen Peroxide. 4. Observe for an immediate pink color.
8. **Utility of Serological Tests:** They allow for the identification of the species of origin (human vs. animal) and the specific type of body fluid present.
9. **Esterase D Typing:** A polymorphic enzyme system found in red blood cells used as a genetic marker to narrow down the pool of potential donors in personal identification.
10. **Iso-enzyme Detection:** Primarily detected in the lab using **Electrophoresis**, where enzymes are separated based on their molecular charge and size.
11. **Polymorphic Proteins:** Proteins that exist in several different forms (variants) within a population; they are forensically important because they provide a "biochemical fingerprint" for individualized profiling.
12. **Screening Test at Scene: Kastle-Mayer (KM) test** or **Luminol** (for latent stains).

Part B: Short Essay Questions

Answer any 5. Each carries 4 marks.

13. Bloodstain Pattern Analysis (BPA): Analysis of the size, shape, and distribution of stains to reconstruct events, such as the direction of force, number of blows, and position of the victim.

14. Microscopic Spermatozoa Examination: Use of the **Christmas Tree Stain** (Picroindigocarmine and Nuclear Fast Red) to visualize the green-tailed, red-headed sperm cells under high magnification.

15. Collection & Preservation: Biological fluids must be air-dried, packaged in breathable paper containers, and stored in cool, dry conditions to prevent DNA degradation.

16. Affinity vs. Avidity: Affinity is the strength of a single interaction between an antigen's epitope and an antibody's binding site. **Avidity** is the overall strength of binding between a multivalent antibody and antigen.

17. Fresh Blood Grouping: Description of the **Forward Grouping** (testing cells with known antisera) and **Reverse Grouping** (testing serum with known reagent cells).

18. Semen Sample Handling: Crucial to prevent the growth of bacteria that consume spermatozoa and to preserve the Acid Phosphatase enzyme or DNA for future profiling.

19. ABO Inheritance: Explained via **Mendelian genetics**; blood groups are determined by three alleles (IA, IB,), where A and B are co-dominant and O is recessive.

Part C: Essay Questions

Answer any 1. Carries 10 marks.

20. HLA Typing:

Paternity: Since HLA markers are highly polymorphic and inherited from parents, they can exclude a putative father with high statistical certainty.

Criminal: HLA antigens on white blood cells can be used to link biological tissue (like skin or blood) to a specific individual.

21. Presumptive & Confirmatory Blood Tests:

Kastle-Mayer (Presumptive): Based on the peroxidase-like activity of hemoglobin. Highly sensitive but prone to false positives from vegetable peroxidases.

Takayama Test (Confirmatory): Heating a bloodstain with a reagent containing pyridine and glucose to form salmon-pink, pyridine-haemochromogen crystals, which are viewed under a microscope.

Scheme of Evaluation

Section	Marking Criteria	Marks
Part A	**2 Marks** for a complete, accurate definition or identification. **1 Mark** for partially correct or overly brief answers.	20

Section	Marking Criteria	Marks
Part B	**2 Marks** for the core procedure/principle. **1 Mark** for forensic application. **1 Mark** for clarity/technical terminology.	20
Part C	**3 Marks** for introduction/definitions. **5 Marks** for detailed procedural/scientific explanation. **2 Marks** for significance/concluding analysis.	10
Total		**50**